

NEXA

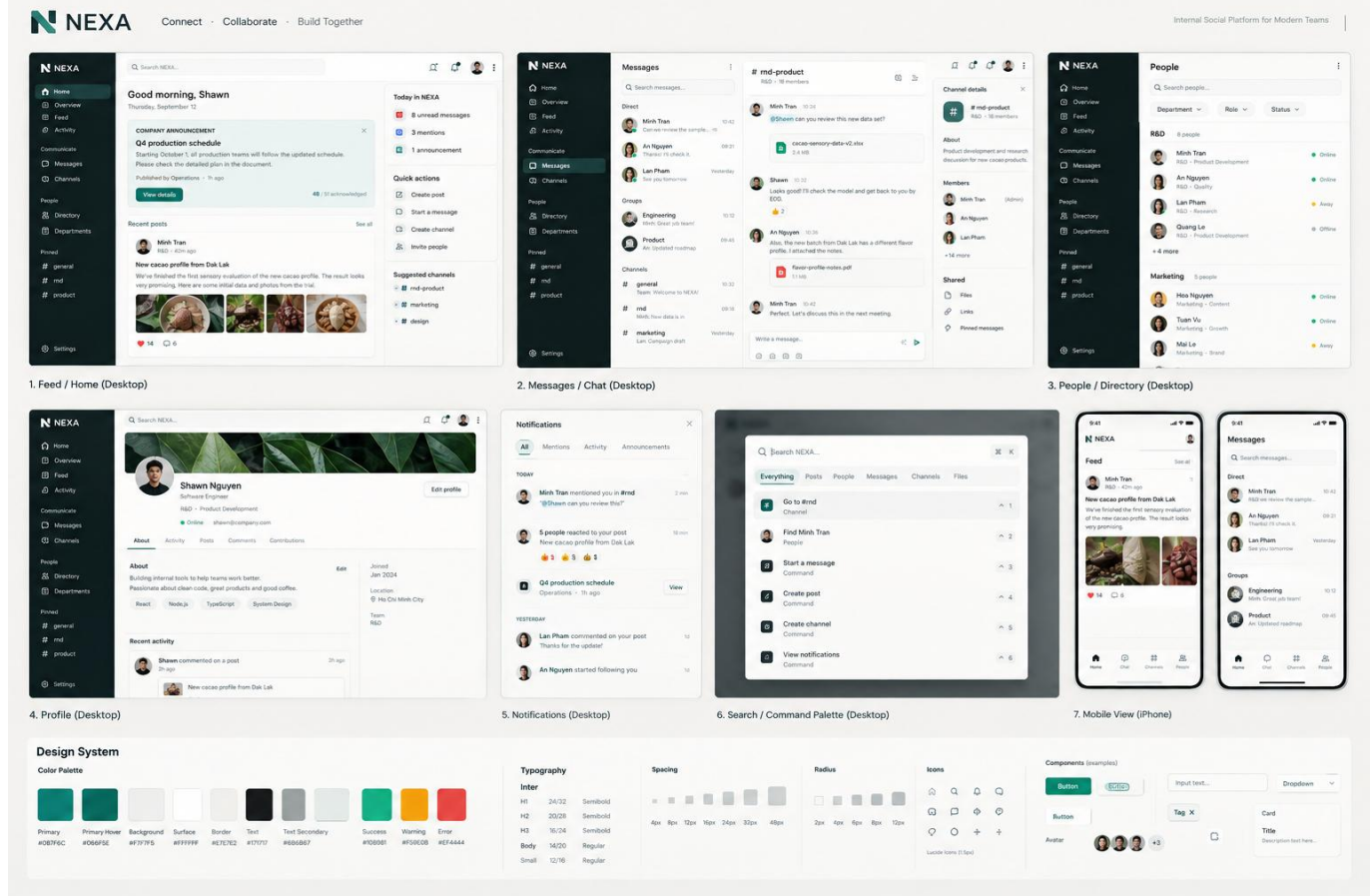
v1 Frontend & Design System Specification

Workplace Social Network / Internal Communication Platform

Purpose: a source of truth for NEXA's frontend UX, visual language, component system, responsive behavior, realtime UI, and implementation structure.

Design direction: professional, calm, information-dense, human, and intentionally restrained. NEXA must feel like a real workplace product, not a generic AI-generated SaaS template.

Document	Role
01 — Architecture Specification	Backend, tenancy, APIs, infrastructure
02 — Edge Cases & Risk Register	Failure modes, security, race conditions, recovery
03 — Frontend & Design System Specification	UX, UI, components, screens, responsive behavior



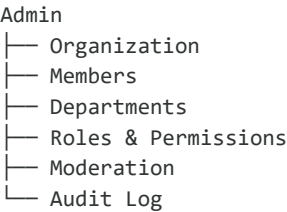
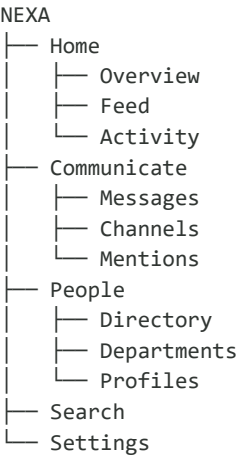
1. Product UX Principles

- Work before decoration — the content/task area gets the strongest visual emphasis.
- Structure should be felt, not seen — use spacing, alignment and hierarchy before heavy cards/borders.
- High information density, low visual noise.
- Context over chrome — navigation recedes after the user reaches the destination.

- Human workplace identity — department/role context is useful where relevant.
- Progressive disclosure — secondary details live in panels, sheets, dialogs and detail views.
- Fast interactions — command palette, keyboard shortcuts, deep links, optimistic reactions and realtime updates.
- No fake complexity — no decorative dashboards, gradients or animation without product purpose.

Reference principle: Linear's recent UI work explicitly emphasizes reduced visual noise, quieter navigation, smaller icon treatment, softer separators and stronger content hierarchy. NEXA should adopt these principles without copying Linear's branding or UI. [\[cite\]](#)[\[turn0search0\]](#)[\[turn0search2\]](#)

2. Information Architecture



Do not make every future feature a top-level nav item. Prefer contextual entry points, pinned items, More menus, search and command palette.

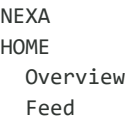
3. App Shell

3.1 Desktop

Global Nav 220–240px	Primary Workspace flexible	Context 300–360px optional
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The context column is optional. Feed can be two-column; chat can be conversation-list + chat; profile can be one main column. Never force three columns everywhere.

3.2 Sidebar



Activity

COMMUNICATE

Messages

Channels

PEOPLE

Directory

PINNED

#general

#rnd

#marketing

Settings

Unread state should be restrained: dot/count and typography, not bright badges everywhere.

4. Visual System

4.1 Light theme

Token	Value	Purpose
background	#F7F7F5	App background
surface	FFFFFF	Primary surface
surface-subtle	#F1F2EF	Hover/secondary surface
text-primary	#171717	Primary text
text-secondary	#6B6B67	Metadata
border	#E7E7E2	Subtle separators
accent	#087F6C	Primary accent
accent-soft	#E4F2EE	Soft accent
danger	#B42318	Destructive/error
warning	#A15C00	Warning

4.2 Dark theme

Token	Value
background	#151614
surface	#191A18
surface-subtle	#20221F
text-primary	#E7E7E2
text-secondary	#A4A7A0
border	#30332F
accent	#35A892

4.3 Typography

Role	Size	Weight
Page title	22–24px	600
Section title	14–16px	600
Body	14px	400
Metadata	12px	400
Navigation	13px	500
Chat message	14px	400

Recommended typeface: Inter or Geist. Keep one family for UI; avoid turning productivity screens into marketing pages.

4.4 Spacing / shape

Property	Rule
Base grid	4px
Common gaps	8 / 12 / 16 / 24px
Radius	6 / 8 / 10–12px
Pills	Only when semantically useful
Shadows	Rare; prefer surface contrast and subtle borders

Avoid the AI-template look of every element being a large rounded card with a shadow.

5. Iconography

- Use one icon family consistently; Lucide is a good baseline.
- 16px navigation icons; ~18px prominent controls.
- Do not add icons to every label.
- Do not use colored icon backgrounds unless color has semantic meaning.
- Keep stroke treatment consistent.

6. Core Screens & Wireframes

Home / Overview

Good morning, Shawn.
Thursday, September 12

COMPANY
Q4 production schedule
Operations · 12 min
[announcement preview]

TODAY IN NEXA
8 unread messages
3 mentions
1 announcement

UX note: A concise workplace briefing, not a dashboard full of metric cards.

Feed

MINH TRAN
R&D · 42 minutes ago

New cacao profile from Dak Lak

We've finished the first sensory evaluation...

[image / attachment]

♥ 14 6 comments

UX note: Posts feel editorial/workplace-oriented; department context is useful.

Announcement

COMPANY ANNOUNCEMENT

Q4 production schedule

Starting October 1, all production teams...

Published by Operations · Sep 12

[Acknowledge] 48 / 51

UX note: Announcements are distinct from normal social posts and may require acknowledgement.

Messages / Chat

Messages	#rnd-product
Direct	R&D · 18 members
Minh	
An	Minh · 10:42
Lan	Can we test the new
Groups	cacao batch tomorrow?
Engineering	
Product	Shawn · 10:44
	Yes. I'll prepare them.
Channels	
#general	Message #rnd-product
#rnd	Send →
#marketing	

UX note: Do not use Messenger-style bubbles. Chat should read like clean workplace communication.

People

People

Search people...

- R&D
- Minh Tran
R&D · Product Development
 - An Nguyen
R&D · Quality

UX note: Prioritize discovery plus department/role context.

Profile

Shawn
Software Engineer
R&D · Product Development

About
Building internal tools...

Activity
Posts · Comments · Contributions

UX note: Avoid Facebook-style cover photos by default.

Notifications

TODAY

MENTIONS

Minh mentioned you in #rnd
"@Shawn can you review this?"
2 min

ACTIVITY

5 people reacted to your post
18 min

COMPANY

Q4 production schedule
Operations · 1h ago

UX note: Group by context/type instead of a flat stream of identical alerts.

Search

Search NEXA

Everything

Posts (8)
Messages (31)
Files (4)
People (2)

Result → deep link to original context

UX note: Search must inherit tenant and authorization scope.

7. Chat UX & Realtime

- Conversation types: Direct, Group, Channel.
- Composer supports text, emoji, attachments and @mentions.
- Message state: sending → sent → failed/retry.
- Client-generated message ID makes retries idempotent.
- Reconnect: authenticate → resubscribe → synchronize missed events.
- Presence is based on active connections, not one boolean.
- Read state synchronizes across devices.
- Deleted messages preserve backend-defined audit semantics.
- Deep links open the conversation and highlight the referenced message.
- Do not yank the user's scroll position when new messages arrive above their current reading point.

State	UI behavior
Sending	Subtle pending state; keep message visible
Sent	Normal presentation
Failed	Visible retry affordance
Offline	Non-blocking connection indicator; preserve drafts
Reconnecting	Show state and resync
New messages	Unread divider/count when user is not at bottom

8. Command Palette

Ctrl/Cmd + K

Search NEXA

Go to #rnd
Find Minh Tran
Start a message
Create post
Create channel
View notifications
Open settings

- Global and permission-aware.
- Search/navigation must not expose inaccessible resources.
- Keyboard shortcuts complement mouse/touch; they never replace them.
- Initial shortcuts: Cmd/Ctrl+K, Esc, Enter, Cmd/Ctrl+Enter where appropriate.

9. Component System

```
packages/ui/  
├─ primitives/  
│   ├── Button / IconButton / Input / Textarea  
│   ├── Avatar / Badge / Tooltip / Popover  
│   └── Dialog / Dropdown / Tabs / Toast  
├─ navigation/  
│   └── Sidebar / Topbar / Breadcrumbs / CommandPalette  
├─ content/  
│   └── Post / Comment / ReactionBar / Attachment / UserCard  
├─ communication/  
│   ├── ConversationList / MessageList / MessageItem  
│   └── MessageComposer / PresenceIndicator  
└─ feedback/  
    ├── EmptyState / LoadingState / ErrorState  
    └── OfflineBanner / ConfirmDialog
```

Layering rule: UI primitives → reusable patterns → domain components → pages. Generic components must not contain business rules.

10. Frontend Architecture

```
apps/web/  
├─ app/  
│   ├── (auth)/  
│   ├── (workspace)/  
│   │   ├── home/  
│   │   ├── feed/  
│   │   ├── messages/  
│   │   ├── channels/  
│   │   ├── people/  
│   │   ├── search/  
│   │   └── settings/  
│   └─ admin/  
├─ features/  
│   ├── auth / organization / feed / chat  
│   └── notifications / search / people
```

```
| components/  
| hooks/  
| stores/  
| lib/  
| styles/
```

```
packages/  
| ui/  
| api-client/  
| realtime/  
| types/  
| config/  
| validation/
```

Recommended implementation: React/Next.js + TypeScript, typed API client, query/cache layer for server state, small local UI state layer, and a dedicated realtime module.

11. State Management

State	Examples	Owner
Server state	Posts, messages, members, notifications	Query/cache layer
Session	User, active organization	Auth/session provider
Realtime	Socket, presence, event stream	Realtime client
UI	Modal, sidebar, active tab	Local state / small UI store
Draft	Message/post composer	Local component state

Keep server cache, realtime state and local UI state conceptually separate.

12. API → UI Contract

```
API  
↓  
API client / validation  
↓  
Query or mutation  
↓  
Domain feature  
↓  
Presentational component  
  
Realtime event  
↓  
Realtime client  
↓  
Domain event handler  
↓  
Cache reconciliation  
↓  
UI
```

- Backend is authoritative for authorization.
- Normalize API errors into safe UI states.
- Optimistic updates must define rollback.
- Prefer cursor pagination where realtime/order/scale requires it.

13. Responsive Design

Viewport	Behavior
≥1280px	Full shell; optional context panel
1024–1279px	Desktop shell; context panel becomes drawer
768–1023px	Reduced sidebar; two-pane where useful
<768px	Mobile navigation; one primary pane; full-screen details

Mobile

NEXA	Bell
Current screen	
Home	Chat Channels People

- Never shrink a desktop three-column layout into mobile.
- Conversation details use a route/sheet/full-screen view.
- Composer must remain usable above the mobile keyboard.
- Bottom navigation contains only high-frequency destinations.

14. Accessibility

- Keyboard navigation for all interactive controls.
- Visible focus states.
- Semantic buttons/links.
- WCAG AA-oriented contrast.
- Never encode meaning by color alone.
- Accessible labels for icon-only buttons.
- Respect reduced-motion preferences.
- Comfortable mobile touch targets.

15. Loading / Empty / Error / Offline

State	Rule
Loading	Skeletons for structured content; avoid spinner-only screens
Empty	Explain what is empty + next useful action
Error	State what failed + retry when meaningful
Permission denied	Do not reveal resource details
Offline	Preserve work and clearly communicate connection
No search results	Show query + useful refinement

No messages yet
Start the conversation with your team.

Upload failed
We couldn't upload this file.
[Try again]

16. Motion & Interaction

- Animation communicates state, continuity or hierarchy.
- Prefer short/subtle transitions.
- Reaction can use a small scale/fade response.
- Panels/dialogs get predictable entry/exit.
- Never animate large content areas without purpose.

17. FE Security & Privacy

- Frontend permission checks are for UX only; backend remains authoritative.
- Do not cache tenant-sensitive data across organization contexts.
- Clear/invalidate tenant-scoped queries and realtime subscriptions on org switch.
- Do not store sensitive message content in localStorage.
- Rich text rendering must follow content sanitization policy.
- File previews must use access-controlled URLs.

18. FE Edge Cases

Case	Expected behavior
User removed from org	Invalidate org-scoped session/cache and move to safe state
Role changed	Refresh effective permissions
Socket disconnect	Reconnect and resync without losing local work
Two tabs	Coordinate unread/presence without corrupting state
Duplicate message retry	Client ID prevents duplicate rendering/persistence
Message fails	Keep it visible with retry
Post deleted while open	Replace with deleted-state UI
Permission revoked	Remove sensitive content and show access state
Organization switched	Reset tenant-scoped queries/subscriptions
Slow network	Preserve drafts; avoid flicker

19. Design Anti-Patterns

- Excessive gradients.
- Glassmorphism as default surface treatment.
- 20–24px radius on every card.
- Huge hero typography inside productivity screens.
- Dashboard made entirely from metric cards.
- Icon for every label.
- Heavy shadows everywhere.
- Decorative AI sparkle motifs when no AI feature exists.
- Unrelated colors used only for variety.
- Animation without a functional reason.

20. FE Development Milestones

Milestone	Scope	Done when
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FE-01 Foundation	Tokens, shell, sidebar, topbar, responsive layout, primitives	All future screens share one shell/language
FE-02 Workplace	Home, Feed, Post, Comments, People, Profile	Core social/workplace loop works end-to-end
FE-03 Communication	Messages, Channels, Composer, Realtime, Presence, Notifications	Chat is reliable and consistent
FE-04 Platform	Search, Command Palette, Admin, Settings, Audit	Product feels complete, not a demo

21. Definition of Done

- Desktop and mobile behavior defined.
- Loading, empty, error, permission and offline states implemented.
- Core flows keyboard-accessible.
- Realtime reconciliation tested where applicable.
- No tenant/user assumptions hard-coded.
- No duplicated shared visual patterns.
- Design tokens used instead of arbitrary raw colors/spacing.
- Realistic data-density visual QA completed.
- Visual hierarchy reviewed before shipping.

22. Recommended Build Order

1. Design tokens
2. App shell
3. Sidebar / Topbar / Command Palette
4. Primitive components
5. Auth + organization context
6. Home / Feed
7. Post / Comment / Reaction
8. People / Profile
9. Conversation list
10. Chat / Composer
11. Realtime synchronization
12. Notifications
13. Search
14. Settings
15. Admin
16. Mobile polish
17. Accessibility + performance pass

Stabilize the shell and design system before building every page. This prevents the frontend from developing multiple incompatible visual languages.

23. AI Coding Agent Rules

- Never invent a new visual pattern when an existing component/pattern fits.
- Do not add gradients, glass effects, giant rounded cards or decorative illustrations without a product reason.
- Before coding a screen, define its shell, hierarchy, states and responsive behavior.
- Use design tokens; do not scatter raw hex values.
- Do not put business rules in generic UI components.
- Do not silently change API contracts to make UI easier.

- New states should update the shared component/domain model rather than becoming one-off hacks.
- Keep domain feature code colocated; keep primitives domain-agnostic.
- Visual QA must include long names, unread counts, many channels, long messages, attachments and permission changes.

24. Product Feel Checklist

Question	Pass condition
Does main content dominate?	Yes; navigation is quieter
Does it look like a template?	No; spacing/hierarchy feel intentional
Can users find communication quickly?	Yes
Can users understand teammates?	Name + department/role where useful
Does density feel overwhelming?	No
Are errors recoverable?	Yes
Does realtime feel trustworthy?	Yes; state is visible
Does mobile feel native to the workflow?	Yes; not merely shrunk

25. Design Reference Notes

The visual direction is original. Linear is used only as a reference for general product-design principles: its 2024 redesign emphasized hierarchy, balance, density, consistent headers/tabs/panels and reduced visual noise; its 2026 refresh further emphasized quieter navigation, reduced icon treatment and softer separators. [\[cite\]](#)[turn0search1](#)[\[turn0search2\]](#)

Linear also documents personalized sidebar behavior, which supports the broader NEXA idea that users can eventually customize frequently used destinations without bloating the main navigation. [\[cite\]](#)[turn0search8](#)

NEXA v1 — quiet enough to work in, distinctive enough to remember.